Akoizo Original Practice - Form 3 - Answer key

Division B | Elastic Launched Gliders | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) If pull distance doubles, about how does stored energy change?

Unit: Launch energy: from stretched elastic to height.
Worked-case data: An elastic with k = 20 N/m is stretched 0.30 m. Half the energy reaches a 6.0 g glider. Find launch speed and ideal height.
Separate investigation observations:
Pull 25 cm: Peak 6 m; flight 19 s.
Pull 35 cm: Peak 9 m; flight 26 s.
Pull 45 cm: Hit 10 m ceiling; flight 15 s.

A. It roughly quadruples

B. It doubles

C. It halves

D. No change

Correct: A

2. (2 points) Which glider mass is legal?

Unit: Rules, scoring, and planning five flights.
Worked-case data: A team with a complete log flies official flights of 22.4, 18.0, 25.1, 9.6, and 24.0 s. What is the final score and tiebreaker?
Separate investigation observations:
Glider A: Flights: 35, 12, 11, 10, 9 s.
Glider B: Flights: 24, 23, 22, 21, 20 s.
Both: Complete flight log.

A. 6.2 g

B. 3.5 g

C. 10.0 g

D. 12.0 g

Correct: A

3. (2 points) Two gliders collide. What may the team do?

Unit: Launch technique and the ceiling.
Worked-case data: In practice, a 35 cm pull reaches 9 m. Competition ceiling is 7.5 m. Estimate a starting pull, assuming height scales with energy.
Separate investigation observations:
Practice room: 12 m ceiling; best pull 40 cm.
Competition room: 6 m ceiling with beams at 5.5 m.
First trim flight: 40 cm pull; glider hit a beam.

A. Add 5 seconds

B. Score the longer flight twice

C. Elect a re-flight after the glider lands

D. Nothing

Correct: C

4. (2 points) Why should the launch roll and glide circle go the same direction?

A. It is required

B. It reduces mass

C. The transition is smoother without reversing the turn

D. It increases launch energy

Correct: C

5. (4 points) Explain why sink rate, not glide ratio alone, predicts time aloft.

answer: Time aloft depends on how fast height is lost; sink rate combines glide ratio and airspeed. (4 points).

6. (4 points) How can peak height be estimated consistently?

Unit: The flight log and competition day.
Worked-case data: A team has five official flights in 6 minutes. After a trim flight, it plans official flights. Its first three are 21, 23, and 8 s (a beam hit). How should it use the remaining two?
Separate investigation observations:
Flight 4: Peak 9 m, time 12 s.
Flight 5: Peak 4 m, time 13 s.
Flight 6: Peak 9 m, time 24 s after reducing rudder offset.

answer: Compare with known heights like the ceiling or wall marks, or use video with a reference object, the same way each time. (4 points).

7. (4 points) Why design the nose early?

Unit: Construction: light, stiff, and legal.
Worked-case data: A glider weighs 3.4 g and its CG is behind the target. What should the team do?
Separate investigation observations:
Flights 1-4: 22-24 s each.
Flight 5: Hard landing on nose.
Flights 6-8: 9-12 s; glider veers right at launch.

answer: Its shape must pass the blunt nose check, and its mass strongly affects the CG. (4 points).

8. (4 points) Why must wings be stiff at launch?

Unit: Launch energy: from stretched elastic to height.
Worked-case data: An elastic with k = 20 N/m is stretched 0.30 m. Half the energy reaches a 6.0 g glider. Find launch speed and ideal height.
Separate investigation observations:
Pull 25 cm: Peak 6 m; flight 19 s.
Pull 35 cm: Peak 9 m; flight 26 s.
Pull 45 cm: Hit 10 m ceiling; flight 15 s.

answer: At launch speed, flexible wings flutter or bend, wasting energy and sending the glider off course. (4 points).

9. (4 points) How is a tie broken?

Unit: Rules, scoring, and planning five flights.
Worked-case data: A team with a complete log flies official flights of 22.4, 18.0, 25.1, 9.6, and 24.0 s. What is the final score and tiebreaker?
Separate investigation observations:
Glider A: Flights: 35, 12, 11, 10, 9 s.
Glider B: Flights: 24, 23, 22, 21, 20 s.
Both: Complete flight log.

answer: By the longest non-scored official flight score. (4 points).

10. (6 points) Solve this case and explain your reasoning: In practice, a 35 cm pull reaches 9 m. Competition ceiling is 7.5 m. Estimate a starting pull, assuming height scales with energy.

reasoning: Height □ energy □ pull^2, so pull □ sqrt height.
New pull about 35 x sqrt (7.0 ÷ 9) about 35 x 0.88 about 31 cm, aiming 0.5 m below the ceiling.
Test with a trim flight and adjust. (3 points).

conclusion: Start near 31 cm, then fine-tune with a trim flight. (3 points).

11. (6 points) Solve this case and explain your reasoning: A glider climbs well but loops over at the top and then dives. Suggest a fix.

reasoning: Looping means the nose-up moment at launch speed is too strong for a straight launch.
Launch with more bank so the climb curves to the side.
If it still loops, reduce decalage slightly and retest the hand glide. (3 points).

conclusion: Bank the launch more, then reduce decalage if needed. (3 points).

12. (2 points) Which glider has the longer glide?

Supplied synthetic observations:
Glider A: Peak 10 m; sink 0.55 m/s.
Glider B: Peak 8 m; sink 0.35 m/s.
Question: Which glides longer, ignoring climb time?

A. Glider A

B. Equal

C. Glider B

Correct: C

13. (2 points) What does the log show about Flight 4?

Supplied synthetic observations:
Flight 4: Peak 9 m, time 12 s.
Flight 5: Peak 4 m, time 13 s.
Flight 6: Peak 9 m, time 24 s after reducing rudder offset.

A. The elastic was worn out

B. It had a good launch but a poor glide trim

C. It had a poor launch

Correct: B

14. (2 points) Define “Sheet balsa wing” in the context of Elastic Launched Gliders.

answer: Solid balsa wing shaped to an airfoil. (2 points).

15. (2 points) Define “Stiffness (k)” in the context of Elastic Launched Gliders.

answer: Force per unit stretch. (2 points).

16. (2 points) Define “Final score” in the context of Elastic Launched Gliders.

answer: Sum of the three best flight scores. (2 points).

17. (2 points) Define “Re-flight” in the context of Elastic Launched Gliders.

answer: Repeat allowed after a collision. (2 points).

18. (2 points) Define “Banked launch” in the context of Elastic Launched Gliders.

answer: Launch with the wings tilted so the climb curves. (2 points).